

# Recap and Project Discussion

COMP 388-002/488-002 Biometrics

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# Today's Attendance

**Please fill out the form**

[forms.gle/1BneyBjHJu9RkqTM6](https://forms.gle/1BneyBjHJu9RkqTM6)



# Recap, midterm exam

## **Biometric Recognition Fundamentals**

What is Biometrics (biometric recognition)?

What are the types of biometric recognition?

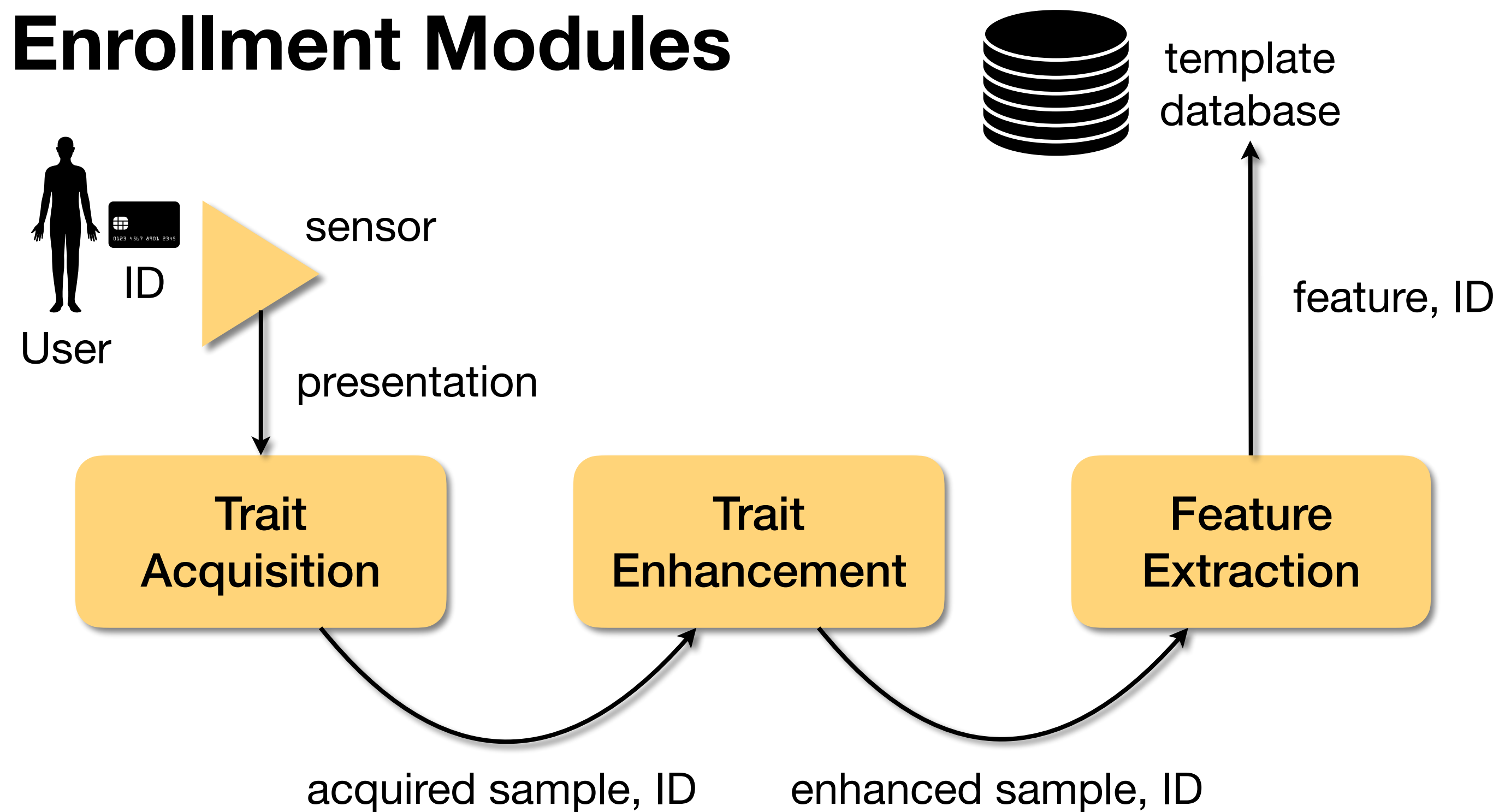
What are the pros and cons of each type?

What is a biometric system?

What are the typical modules of a biometric system?

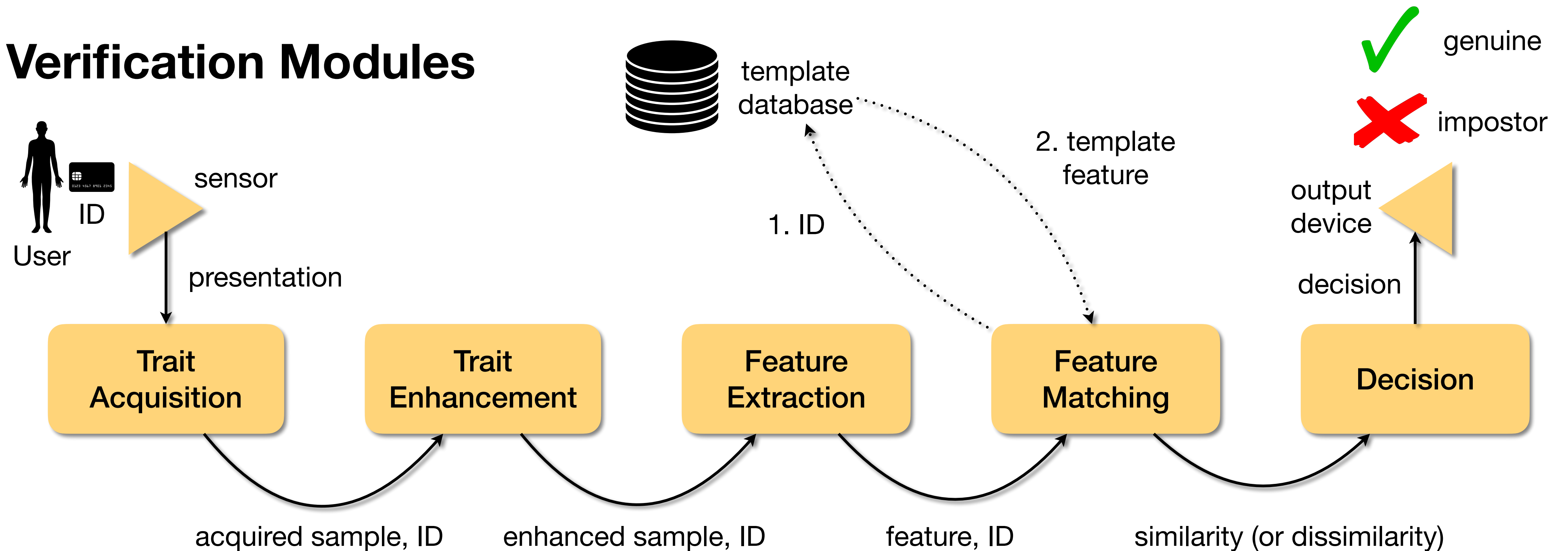
# Biometric Systems

## Enrollment Modules



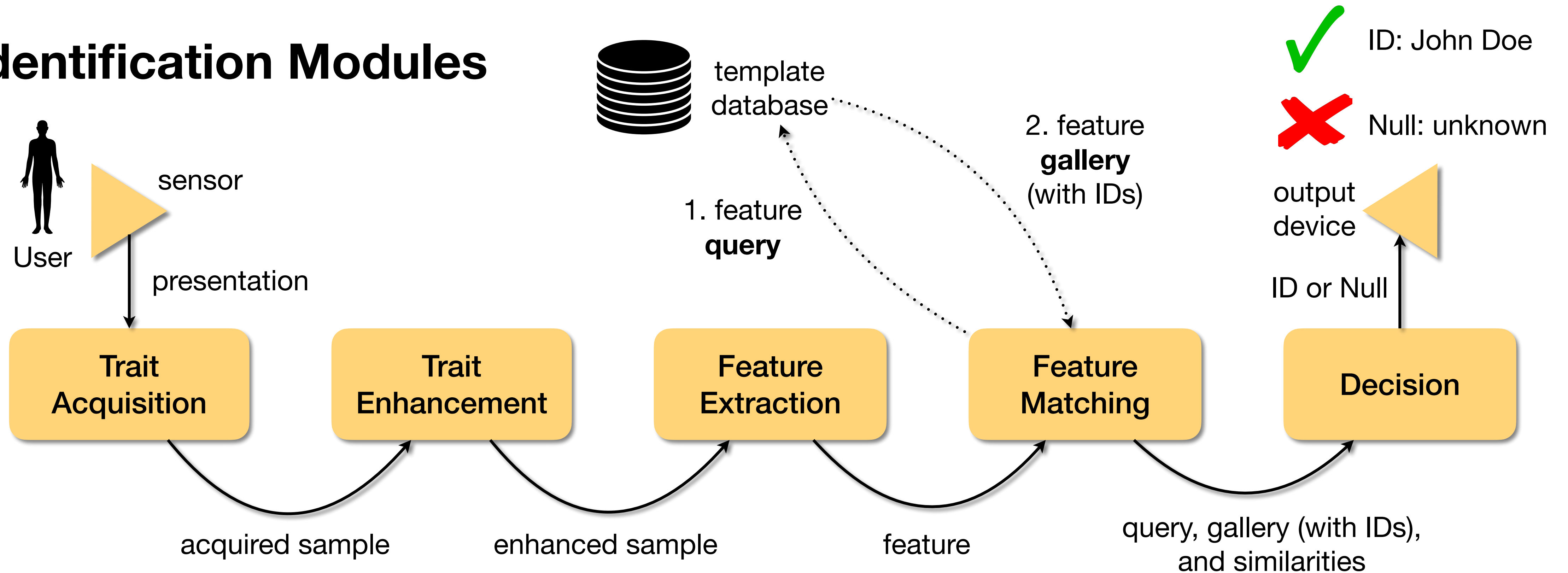
# Biometric Systems

## Verification Modules



# Biometric Systems

## Identification Modules



# Recap, midterm exam

## Utility

Why do we use Biometrics?

What are the pros and cons of using Biometrics?

Why do we use computers in Biometrics?

## Traits

What is a biometric trait?

What is not a biometric trait?

What should one consider when choosing traits?

# Recap, midterm exam

## **Biometric Identification**

What is open set recognition?

Why do we need open set recognition in Biometrics?

How does one distinguish between an open set and a closed set system?

How can one make a recognition system an open set?

# Recap, midterm exam

## Matching

What is the typical output of a Biometric system?

Score versus *decision* output: utility versus security.

Types of score: *distance* versus *similarity*.

Decision threshold: concept and setup.

# Recap, midterm exam

## Metrics

How do we compare two or more systems?

Genuine versus impostor pairs

FMR, FNMR, TMR, TNMR, and EER

d-prime and AUC

Similarity versus distance-based scores

# Recap, midterm exam

## **Ethical aspects**

Misuse, abuse, and bias

Function creep

Covert deployment

## **Errors and attacks**

Denial of access, intrusion, and repudiation

Impersonation, obfuscation, spoofing, hill climbing, and master keys

# Recap, midterm exam

## **Fingerprints**

Epigenetic traits

Ridges and valleys

Level-1, level-2 and level-3 features

*Examples and utility*

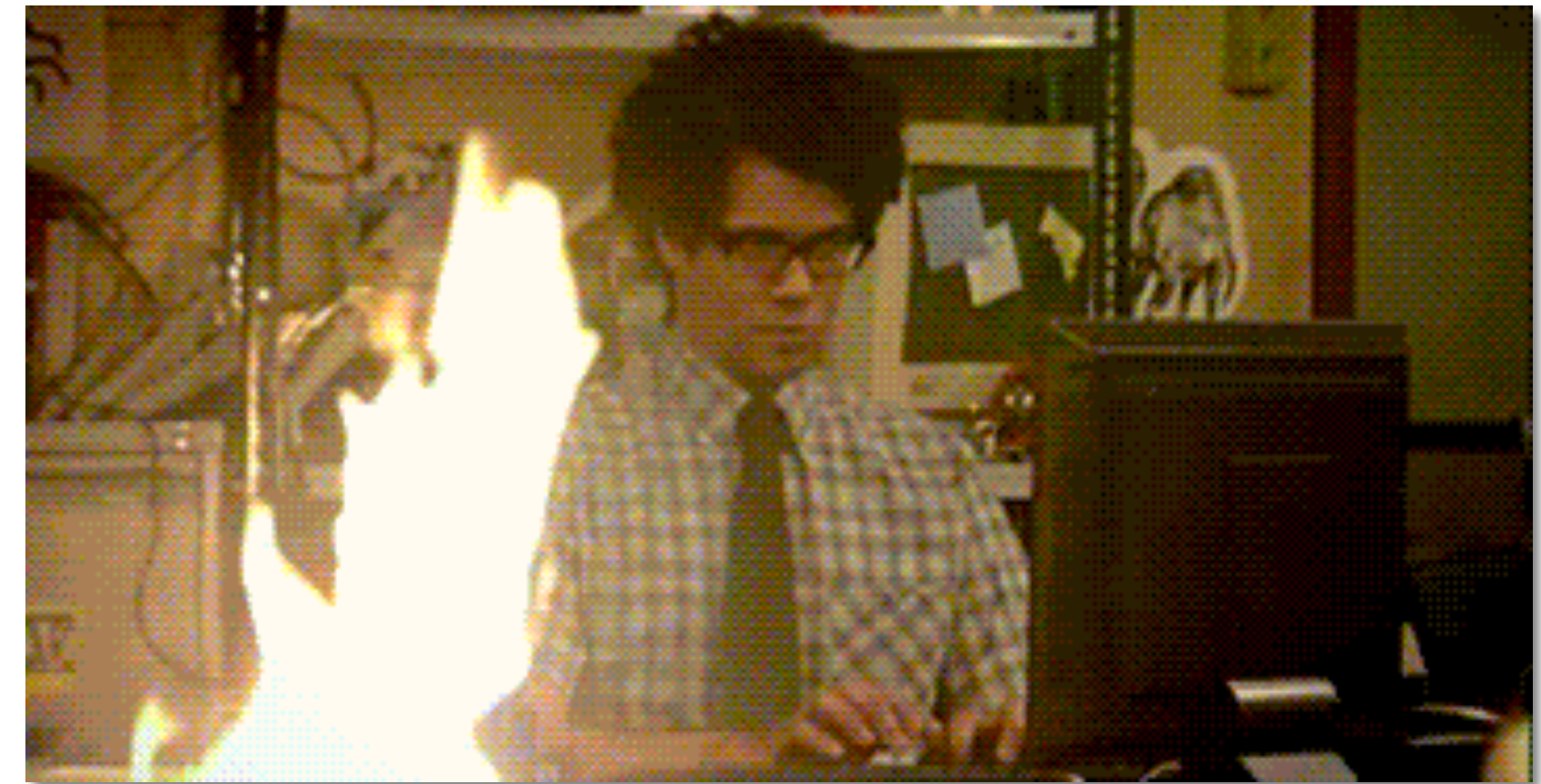
Types of sensors and their resolutions

*Capacitive, optical, pressure,  
thermal, ultrasound, and touchless*

# Assignment 2

## **Fingerprint recognition**

Open today and due on 10/20.



# What's Next?

## Midterm exam

In class on Thursday

On paper

One-page cheat sheet is allowed

